

# From One Objective to Measured Boundaries: Channel–Code Coupled OFDM Reception on Measured Underwater Channels

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**Abstract**—Coupled underwater-acoustic OFDM receivers such as JUNA fold code structure into channel estimation. This paper unifies the theory and the measured evidence of that program around one organizing hypothesis: coupling pays only where the pilot budget leaves channel information unresolved — an *estimation deficit* — and where initial decisions are reliable enough to feed back; the gain is a property of the operating point, not of a channel label. One hybrid generative–discriminative objective generates every receiver used here as an exact specialization, an exact conditional update, or a declared engineering approximation, so the experiments compare declared instruments rather than ad-hoc algorithms. On measured replay channels the prediction holds in direction. At 71 matched waveforms, coupling gains a median 15.9% effective rate on a harbour channel — falling to 1.8% when each receiver is instead allowed its own confirmed best configuration — and 0.0% on a tuned deep-water channel. A Holm-corrected paired test on unseen frames promotes a coupled receiver on four of nine harbour paths and none of nine deep-water paths. At one deliberately low-rate frozen waveform, the same deep-water family flips to 225/7/8 in favour of adaptive coupling (mean effective rate 38.0 to 207.6 bit/s; descriptive, correlated paths). A preregistered factorial then tests the sharpest mechanism corollary — that sparser pilots should enlarge the value of a two-block coding horizon — and does *not* establish it: the interaction signs differ by path and all three 95% intervals include zero. The same experiment isolates the actual bottleneck: a genie shows large code-derived headroom on the harbour and deep-water paths, yet real belief-propagation feedback captures less than 0.2% of it, and conditioned joint updates change hundreds of internal trajectories with zero accepted endpoint change. A third family bounds the account: within one deployment family, coupling helps two captures, ties one, and regresses three (descriptive; one noise seed and three frames per path), and no tested receiver Pareto-beats OFDM + FEC. The defensible synthesis is a design law: coding horizon, physical pilots, feedback quality, and final acceptance are separate levers, and the present limit of the program is converting uncertain decoder beliefs into trustworthy continuous-state updates.

**Index Terms**—Underwater acoustic communication, OFDM, intercarrier interference, partial-FFT, LDPC, joint estimation and decoding, pilot allocation, decision feedback, receiver verification.

## I. INTRODUCTION

**A**N OFDM receiver is simple when the cyclic prefix contains the delay spread and the channel is nearly constant over a useful symbol. Residual time scaling and within-

Technical report, Acoustic Research Laboratory. The receiver mapping is synchronized with the executable JunaCore implementation; every reported number is regenerated from checked-in sweep data pinned by executable evidence contracts.

symbol variation break the second assumption: a symbol transmitted on carrier  $q$  contributes to several observed carriers  $k$ , and different temporal portions of the FFT interval carry different mixtures of the same symbols. A free intercarrier-interference (ICI) model can fit this behavior but can also absorb almost any candidate data vector; a diagonal receiver avoids that ambiguity but discards observation structure. The JUNA family occupies the space between these extremes by combining pilots, partial-FFT views, decoder information, and constrained ICI models.

“Joint” means two different things in this program, and conflating them has historically made the evidence hard to read:

- 1) *joint code horizon*: one LDPC Tanner graph spans several transmitted OFDM blocks instead of one graph per block; and
- 2) *joint continuous refinement*: the receiver updates channel, combiner, or soft-symbol variables using decoder feedback.

This paper keeps the two mechanisms separate throughout, assigns each its own experiments, and reports where each one pays, where it is inert, and where it regresses.

The paper is organized as a single argument. Section II states the one-shot model and its failure boundary. Section III states the estimation-deficit hypothesis — the one organizing claim every later section tests. Section IV derives the instrument family from one declared objective, so that receiver comparisons are comparisons between declared approximations. Section V fixes channels, replay protocols, metrics, and the evidence map. Sections VI–VIII report the three outcome tests: isolation at matched waveforms, significance on unseen frames, and breadth across 267 paths. Section IX isolates iteration depth as a separate lever. Section X reports the preregistered mechanism experiment, including its null primary endpoint and the measured feedback bottleneck. Section XI reports the boundary family. Section XII states what is and is not established, Section XIII bounds the evidence, and Section XIV closes the account.

**Notation.** Throughout,  $K$  denotes the ICI band half-width of the receiver model and nothing else. The coding horizon — the number of OFDM blocks one LDPC codeword spans — is written  $H_c$ ; raw-data labels  $K1/K2$  in archived Experiment B artifacts mean  $H_c = 1$  and  $H_c = 2$  and never the ICI width.  $M$  denotes the number of partial-FFT temporal views. “Frame-wide” describes a codeword spanning all blocks of a frame; it is a configuration of a receiver, not a separate algorithm.

The optimized logits of the coupled receivers relate to decoder log-likelihood ratios by  $z = -L$ , so soft symbols  $s(z)$  and confidence weights  $\tanh(L/2)$  refer to the same posterior with opposite sign convention.

## II. ONE-SHOT RECEPTION AND THE FAILURE BOUNDARY

### A. Observation model

For one OFDM block with  $N$  carriers, cyclic prefix  $N_{\text{CP}}$ , and transmitted frequency-domain symbols  $x \in \mathbb{C}^N$ , the received frequency-domain observation is

$$y = Cx + n, \quad (1)$$

where  $C$  is the channel coupling matrix and  $n$  is noise. When the channel is constant over the symbol and the prefix contains the delay spread,  $C$  is diagonal and one complex tap per carrier suffices. Residual Doppler and within-symbol variation populate off-diagonal entries; the receiver model constrains them to a band:  $C_{kq} = 0$  for  $|k - q| > K$ , with the coefficients tied across a small number of frequency bands (band-tied parameterization). The neighborhood  $\mathcal{N}_K(k) = \{q : |k - q| \leq K\}$  defines which transmitted symbols the receiver believes can reach observed carrier  $k$ .

A partial-FFT front end splits the symbol interval into  $M$  temporal parts and computes one FFT per part, yielding views  $y^{(1)}, \dots, y^{(M)}$ . Each view sees a shorter time span, so within-symbol variation is closer to constant per view; a per-band combiner  $W$  then mixes the views. Standard OFDM is the  $M = 1$  special case. Among the one-shot receivers, OFDM + FEC estimates  $C$  from pilots, whereas pilot-only Partial-FFT estimates the band-tied combiner  $W$  without forming  $C$ ; both then equalize and decode once.

Figure 1 locates the distinct one-shot, posterior-anchor, and joint-refinement operations in the shared signal chain.

### B. The failure boundary

Standard OFDM is an exact specialization of the diagonal model (1) under pilot-only channel fitting, and pilot-only Partial-FFT is the exact conditional ridge minimizer of its pilot-calibration loss [1]. Their failure boundary is therefore informational, not algorithmic: when the number of active taps and the per-tap Doppler rate exceed what the pilot budget can resolve within a coherence interval, the pilot-only estimate of  $C$  goes stale within a frame and no amount of per-carrier equalization recovers the loss. Exactly at that boundary, the decoder still holds unused information about the transmitted symbols — which motivates coupling, and the hypothesis of the next section.

## III. THE ESTIMATION-DEFICIT HYPOTHESIS

The value of folding code structure into channel estimation is governed by the channel knowledge that the code can supply beyond what the pilot comb already resolves — informally, the conditional mutual information between the transmitted symbols and the channel given the pilots [2]. We call this the *estimation deficit* of an operating point.

The hypothesis makes directional predictions:

- P1. Where the deficit is large and initial decisions are reliable, tentatively decoded symbols act as additional pilots and coupling pays.
- P2. Where pilots already identify the channel, per-carrier likelihoods are close to a sufficient statistic and coupling gains approximately nothing.
- P3. The deficit is set by the waveform, pilot budget, and channel dynamics together — an operating-point property. The same physical channel family can sit on either side of the boundary under different waveforms.

Two things the hypothesis does *not* claim: it does not make an approximate implementation automatically non-regressing (an implemented coupled receiver may lose where its front end, model, optimization, or selection rule is approximate), and it does not predict that a channel *label* fixes the sign of the gain. Sections VI, VII, and VIII test the predictions on outcomes; Section X tests the mechanism under preregistered control.

## IV. THE INSTRUMENT FAMILY: ONE OBJECTIVE, DECLARED DESCENDANTS

Every receiver in this program descends from one declared root: the hybrid generative–discriminative variational objective  $J(C, W, z)$  over the channel response, the partial-FFT combiner, and relaxed bit logits, or, for Fully Coupled, its generative MAP parent with a physics prior. Code constraints enter through decoder beliefs [1]. The hybrid root is a declared design objective, not a MAP posterior; that distinction permits the accounting summarized in Table I. Ideal sum-product BP fixed points are stationary points of the complete Bethe variational problem, with code constraints entering through the Bethe code functional; they are not guaranteed global minimizers. The deployed normalized min-sum decoder is a marked approximate projection [1].

The campaigns below use rosters of different sizes drawn from Table I plus two Yellow-only diagnostic arms (Turbo-MAP and frame-wide band-RLS). Experiment A compares six receivers (the two one-shot receivers, JUNA-Lite, and ungated, CRC-gated, and conditioned joint refinement); Experiment B records eight rows per cell (adding the frozen- and genie-feedback  $(C, z)$  controls). The Yellow stress test drops the ungated  $(C, z)$  arm and reports nine by adding Turbo-MAP, a profiled gradient, a frame-level  $(C, z)$  gradient, and frame-wide band-RLS. These larger-roster entries are diagnostic arms, not new mechanisms; Fully Coupled is not tested in these campaigns.

## V. CHANNELS, PROTOCOLS, AND THE EVIDENCE MAP

### A. Measured replay channels

Three measured channel families are used, replayed through a calibrated passband adapter with additive noise quoted as added-noise SNR. *Red* is a Singapore harbour capture: roughly 244 active taps spanning 32.8 ms to the  $-30$  dB point, with fast per-tap Doppler. *Blue* is a North Atlantic deep-water capture: 15 taps spanning 1.43 ms with essentially frozen tap gains plus a bulk Doppler term removed exactly by a  $\pm 5000$  ppm resampling search. *Yellow* is a third-family deployment set of six captures replayed over receiver lanes under a native

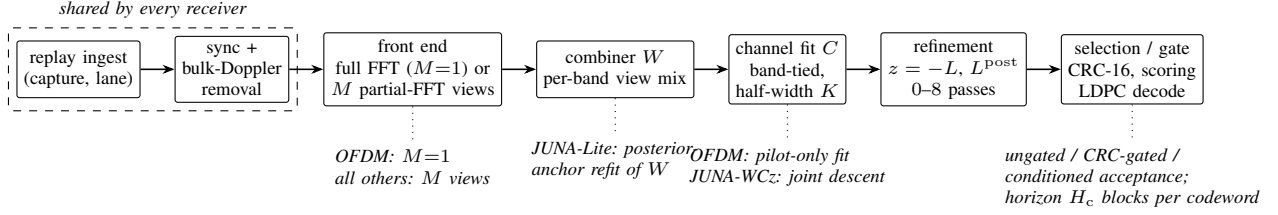


Fig. 1. The receiver signal chain. Every variant shares the ingest and synchronization front end and diverges at exactly three stages: the front end (number of partial-FFT views  $M$ ), the continuous-state estimation (what information updates  $C$ ,  $W$ , and the logits  $z$ ), and the refinement/selection stage (feedback passes, acceptance rule, and coding horizon  $H_c$ ). The mathematical objects named in Section IV live at the indicated stages.

TABLE I  
THE INSTRUMENT FAMILY. STATUS ENTRIES ARE STRUCTURAL CLAIMS MADE BEFORE ANY DATA IS CONSULTED. “FRAME-WIDE” ( $H_c > 1$ , UP TO ALL BLOCKS OF A FRAME) IS A CONFIGURATION AVAILABLE TO EACH RECEIVER, NOT A SEPARATE ROW.

| Receiver                                                                                            | Update                                                                                                                   | Status                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OFDM + FEC                                                                                          | pilot-only $C$ , $M=1$ , decode once                                                                                     | exact specialization                                                                                                                                                         |
| Partial-FFT + FEC                                                                                   | pilot-only band-tied $W$ over $M$ views; $C$ never formed                                                                | exact conditional ridge minimizer                                                                                                                                            |
| JUNA-Lite                                                                                           | ridge refit of the band combiner $W$ anchored on frozen decoder posteriors                                               | exact conditional minimizer; bounded confidence weights and candidate selection are engineering approximations related to, but not a restriction of, the canonical objective |
| JUNA-Wz and profiled ( $C$ , $z$ ) family (frozen-, real- (ungated), and genie-feedback; CRC-gated) | confidence-weighted proximal loss over $W$ , $z$ , or a constrained frame ( $C$ , $z$ ) gradient, at fixed anchor values |                                                                                                                                                                              |
| JUNA-WCz (conditioned joint ( $C$ , $W$ , $z$ ))                                                    | descent of the canonical objective, band-tied parameterization                                                           | declared approximation; joint identifiability requires a bilinear rank condition                                                                                             |
| Fully Coupled                                                                                       | frame-stateful frozen fixed-multipath member; uncertainty-aware linear updates alternating with per-packet sum-product   | declared approximation                                                                                                                                                       |

two-LFM synchronization protocol; its configuration-selection pipeline differs from the Red/Blue campaigns and it is never pooled with them.

### B. Metrics

Effective payload rate is CRC-exact delivered payload divided by waveform airtime. Payload BER is counted over scored payload bits; exact-frame PSR is the fraction of frames decoded without error. Experiment A additionally reports the correct-bit throughput

$$R_{\text{correct}} = \frac{FB - E}{FT} \quad (2)$$

for  $F$  frames of  $B$  payload bits,  $E$  total payload-bit errors, and duration  $T$ ; this is a descriptive soft-success measure that credits erroneous bits and is not delivered application goodput.

### C. The evidence map

Table II is the single-source index of every campaign in this paper. Each headline number is listed with its mandatory qualifier; quoting a headline without its qualifier misstates the evidence.

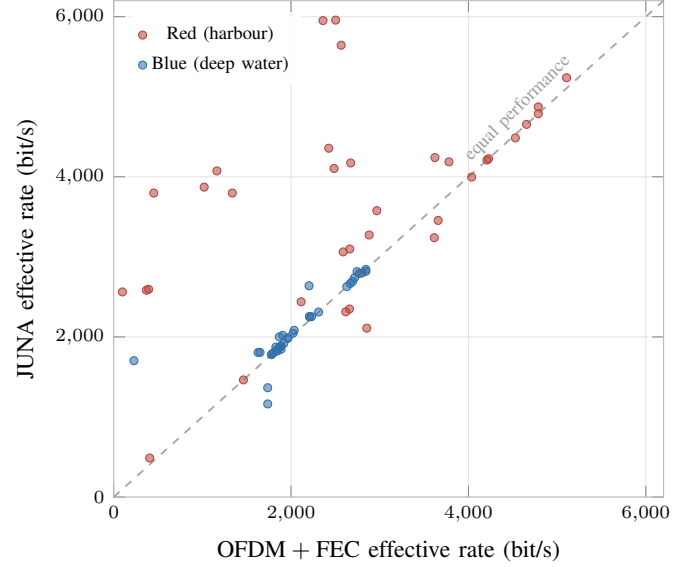


Fig. 2. Effective payload rate of JUNA against OFDM + FEC at 71 matched waveforms on measured replay channels at 20 dB. Blue (deep water): median gain 0.0%, split 17/5/13, including one large rescue and two repeatable regressions. Red (harbour): median +15.9%, mean +160.9%; the median-mean gap reflects a minority of waveforms at which pilot-only estimation fails outright while code-aided estimation still closes the link. The Red median falls to 1.8% when each receiver is allowed its own confirmed best configuration.

## VI. TEST 1: ISOLATING THE RECEIVER AT MATCHED WAVEFORMS

The configuration searches that produced the Red and Blue finalists rank waveform and receiver jointly, so a table of per-path winners cannot separate the two. We therefore extract every case in which OFDM + FEC and at least one JUNA variant were confirmed at the *identical* waveform within the same pinned campaign — all seven swept fields (FFT length, cyclic prefix, LDPC rate, both pilot spacings, check degree, coding horizon) agreeing — yielding 71 matched waveforms: 36 on Red and 35 on Blue, at 20 dB with one-second CRC-16 frames. Both receivers decode the same transmitted frames through the same replayed channel at the same noise realization, so the two axes of Fig. 2 differ only in which receiver produced the estimate.

On Blue the median gain from coupling is 0.0%: better on 17 of 35, worse on 5, exactly tied on 13. On Red the cloud lifts from the diagonal: median +15.9%, 23 of

TABLE II  
EVIDENCE MAP. EVERY CAMPAIGN, ITS DESIGN STATUS, ITS HEADLINE, AND THE QUALIFIER THAT MUST TRAVEL WITH THE HEADLINE.

| Campaign                        | Scale                                                                               | Design status                                | Headline                                                                                                                            | Attached qualifier                                                                                                    |
|---------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Matched waveforms (Sec. VI)     | 71 waveforms (36 Red, 35 Blue), 20 dB                                               | paired at identical waveform                 | Red median +15.9% (23/7/6); Blue 0.0% (17/5/13)                                                                                     | Red median falls to 1.8% under own-best configurations; two repeatable Blue regressions ( $\approx 21\%$ , $33\%$ )   |
| Paired receiver call (Sec. VII) | 180 unseen frames per path, 18 paths                                                | paired, Holm-corrected McNemar               | 4/9 Red paths promoted; 0/9 Blue                                                                                                    | single co-selected waveform handicaps the non-choosing receiver; rate ratios are not effect sizes                     |
| Blue frozen family (Sec. VIII)  | 240 capture-lane paths, 9 frames each                                               | descriptive, correlated                      | 225/7/8; mean rate 38.0 $\rightarrow$ 207.6 bit/s                                                                                   | not powered; captures and lanes correlated; waveform is not either receiver's optimum                                 |
| Experiment A (Sec. VIII)        | 267 paths, one frozen waveform, 14,418 outcomes                                     | descriptive                                  | JUNA-Lite 259/6/2 on correct-bit throughput                                                                                         | metric credits erroneous bits; 240 Blue paths correlated                                                              |
| Iteration depth (Sec. IX)       | one path, 16 SNRs, passes 1–8                                                       | diagnostic                                   | 40 improve / 28 worsen / 44 tie                                                                                                     | single path and geometry; forced passes, not the deployed stopping rule                                               |
| Experiment B (Sec. X)           | preregistered factorial, 51,840 outcomes                                            | preregistered                                | pilot-scarcity interaction <i>not established</i> (all three 95% CIs include 0); real BP feedback captures < 0.2% of genie headroom | genie is an optimistic bound; pilot arms domain-separated, so cross-arm contrasts are unpaired                        |
| $P = 1$ controls (Sec. XI-A)    | Blue-1 and Yellow-1, 16 SNRs, 1,350 frames per receiver and SNR within each channel | receiver-matched within channel, descriptive | five-receiver BER/SNR curves with full-FFT-equivalent front-end observations                                                        | channel-specific waveforms; separate replay campaigns, not a paired cross-channel comparison; no confidence intervals |
| Yellow stress (Sec. XI)         | 18 paths, SNR 0:2:30 dB                                                             | descriptive, one noise seed                  | OFDM + FEC lowest equal-path BER at every SNR; refiners collapse onto JUNA-Lite                                                     | three frames per path and SNR; conservative bandwidth                                                                 |

36 favouring coupling, mean +160.9%. The dispersion is strongly asymmetric because a small number of configurations dominate: at the most extreme Red waveform, OFDM + FEC delivers 98.5 bit/s and JUNA-Lite delivers 2562.2 bit/s. This asymmetry is the deficit account (P1/P2) in one figure: the 32.8 ms rapidly varying Red response exceeds any affordable pilot budget, while most tuned 1.43 ms Blue operating points leave approximately nothing for coupling to recover.

Three qualifiers are part of the result. First, median neutrality on Blue is not a safety result: two repeatable Blue regressions reduce effective rate by approximately 21% and 33% on both confirmation seeds (which share 59 of 60 replay segments, so they are consistent, not independent, replications). Second, the deficit — not the label — governs: Blue also contains a rescue waveform at which OFDM + FEC delivers 228.5 bit/s against 1703.0 bit/s coupled (P3). Third, holding the waveform fixed denies OFDM + FEC the freedom to retune: when each receiver is allowed its own confirmed best configuration, the Red median gain falls from 15.9% to 1.8%, so coupling and waveform retuning act as partial substitutes over much of the operating space.

## VII. TEST 2: A PAIRED RECEIVER CALL ON UNSEEN FRAMES

Matched waveforms isolate the receiver effect but carry no error bars. For each of the 18 Red/Blue paths, every receiver therefore decoded 180 fresh one-second frames — three seeds spaced wider than the frame count, disjoint from all search seeds — at that path's confirmed best waveform. Per-frame outcomes are genuinely paired and compared with a two-sided exact McNemar test on the discordant frames; the challenger was fixed before the frames were drawn, and significance is Holm-corrected across all eighteen paths.

On Blue no challenger is promoted on any of the nine paths: the largest margins are 3 and 7 discordant frames at raw  $p = 0.25$  and  $p = 0.34$ , and on five paths the two

receivers agree on every frame. On Red a coupled receiver is promoted on four of nine paths, with 80–104 discordant frames won against 0–6 lost and Holm-adjusted  $p$  between  $10^{-22}$  and  $4 \times 10^{-28}$ . The promoted paths are exactly those whose confirmed waveform sits in the long-FFT regime that the harbour delay spread demands. The matched-waveform comparison carries the magnitude; this test carries the significance. Because all receivers are evaluated at a single co-selected waveform, its rate ratios should not be quoted as effect sizes.

## VIII. TEST 3: BREADTH AT DEFICIT-INDUCING OPERATING POINTS

### A. The frozen-waveform Blue family

The deficit account predicts (P3) that the Blue *label* does not fix the sign of the gain: at an operating point that reopens the deficit, the same channels should reward coupling. Twenty Blue captures and twelve receiver lanes are treated as one family — 240 correlated capture-lane paths — and decoded at one common frozen waveform selected in the Blue-1–3 search: QPSK,  $N = 2048$ ,  $N_{CP} = 64$ , rate-1/8 LDPC, outer and inner pilot spacings of 3, check degree 8, one-block coding horizon, 210 payload bits protected by CRC-16 in 0.6527 s. Standard OFDM + frame-wide BP and adaptive JUNA-Lite decode identical frames at 20 dB; three widely separated seeds supply nine paired frames per path; no decode invocation fails.

Across the family (Fig. 3, Table III), adaptive Lite wins effective rate on 225 paths, loses on 7, ties on 8; mean effective rate rises from 38.0 to 207.6 bit/s, the median within-path increase is 178.7 bit/s, equal-path BER falls from 0.0552 to 0.00795, and exact-frame completion rises from 255/2160 to 1394/2160 (1170 Lite-only versus 31 OFDM-only completions). Nineteen of twenty capture means favour Lite. The test is descriptive — correlated measurements, not 240 independent deployments — and its frozen low-rate waveform is

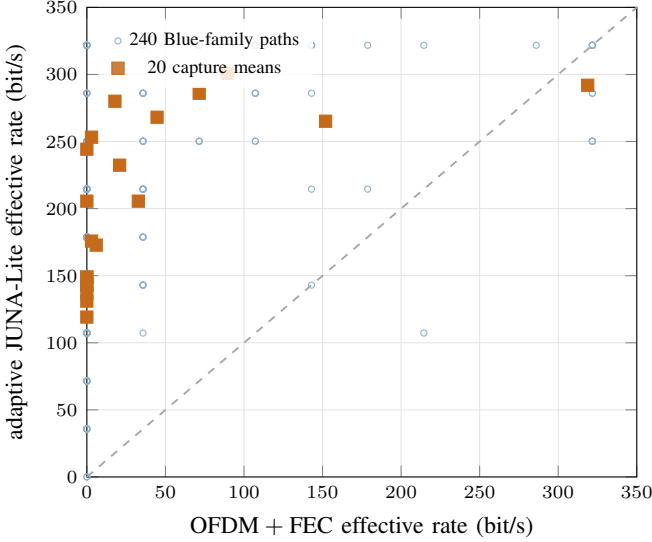


Fig. 3. Frozen-waveform Blue-family test at 20dB. Open circles are all 240 capture-lane paths; squares are the twenty capture means. The dashed diagonal marks equal effective payload rate.

TABLE III  
BLUE FROZEN-WAVEFORM FAMILY EVIDENCE.  $\widetilde{\Delta R}$  IS THE MEDIAN WITHIN-PATH LITE-MINUS-OFDM RATE DIFFERENCE; B/W/T COUNTS PATHS ON WHICH LITE IS BETTER/WORSE/TIED; EXACT REPORTS OFDM VERSUS LITE PAYLOAD-EXACT FRAMES.

| Family      | Paths | OFDM | Lite  | $\widetilde{\Delta R}$ | B/W/T   | Exact                 |
|-------------|-------|------|-------|------------------------|---------|-----------------------|
|             |       |      | bit/s |                        |         |                       |
| Blue family | 240   | 38.0 | 207.6 | 178.7                  | 225/7/8 | 255/2160 vs 1394/2160 |

deliberately not a maximum-goodput setting. Because adaptive Lite includes both pilot-conditioned partial-FFT selection and the posterior-anchor refit, this experiment does not attribute the rescue to either component alone.

#### B. Experiment A: one waveform on all 267 paths

To make the comparison apples-to-apples across all three families, Experiment A freezes one transmitted signal —  $N = 1024$ , no cyclic prefix, rate-1/4 LDPC with check degree 8, pilot spacings 8/10, one codeword spanning nineteen OFDM blocks ( $H_c = 19$ ), 23,552 samples or 4.8234496 s, 7576 payload bits — and replays it on all 267 paths: nine Red, 240 Blue, eighteen Yellow. Six receivers each process nine identical frames per path, 14,418 outcomes in all, scored by correct-bit throughput (2).

All five challengers beat the baseline on every Red and Blue path under this metric (Table IV); Yellow separates them. Figure 4 shows the corresponding path-level comparison. The broad Blue win does not contradict the 0.0% matched median: they are two operating points of the same channels — the matched study tests low-deficit path-tuned configurations, whereas Experiment A imposes one frame-wide-code waveform with no cyclic prefix on every channel, and the narrow Blue baseline band indicates a shared operating-point limitation that all adaptive variants alleviate. CRC gating and

TABLE IV  
EXPERIMENT A CORRECT-BIT-THROUGHPUT BETTER/WORSE/TIED COUNTS RELATIVE TO OFDM + FEC. THIS DESCRIPTIVE METRIC CREDITS ERRONEOUS BITS AND IS NOT DELIVERED APPLICATION GOODPUT.

| Challenger                | Red   | Blue    | Yellow | All      |
|---------------------------|-------|---------|--------|----------|
| Partial-FFT + FEC         | 9/0/0 | 240/0/0 | 6/11/1 | 255/11/1 |
| JUNA-Lite                 | 9/0/0 | 240/0/0 | 10/6/2 | 259/6/2  |
| Ungated ( $C, z$ )        | 9/0/0 | 240/0/0 | 9/4/5  | 258/4/5  |
| CRC-gated ( $C, z$ )      | 9/0/0 | 240/0/0 | 10/6/2 | 259/6/2  |
| Conditioned ( $C, W, z$ ) | 9/0/0 | 240/0/0 | 10/6/2 | 259/6/2  |

conditioned joint selection return JUNA-Lite’s decision pattern, so Experiment A supplies no evidence of an incremental accepted gain from those two mechanisms.

#### IX. A SEPARATE LEVER: ITERATION DEPTH

Coupling gain and iteration gain are distinct empirical questions. To isolate depth, the tuned Red-1 lane-1 geometry is frozen ( $N = 2048$ , zero cyclic prefix, rate-1/2 LDPC, pilot spacings 3/10, check degree 10, four-block frame-wide code, CRC-16) and the posterior-anchor update is forced through exact passes one to eight at each SNR in 0 : 2 : 30 dB, thirty frames per point (146,400 tested payload bits per pass per point), recording the raw decision after each pass so a late regression cannot be hidden by rollback.

Figure 5 shows that the refinement matters principally in the waterfall: at 18 dB the error count falls from 1273 after pass one to 273, 5, and 2 after passes two through four; at 20 dB pass one leaves 233 errors and every later pass is payload-exact; at 22 dB pass one leaves one error; at 24–30 dB all eight passes are exact over the tested bits. Across all sixteen SNRs the 112 adjacent-pass transitions split 40 improve / 28 worsen / 44 tie, and nine of sixteen SNR cells are non-monotonic. Feedback depth can move the link by orders of magnitude, but there is no monotonic iteration guarantee: a practical receiver needs a transactional acceptance rule, not the assumption that more feedback is always safer. This single-path diagnostic motivates the controlled mechanism experiment of the next section.

#### X. THE CONTROLLED MECHANISM TEST: EXPERIMENT B

Experiment B is the preregistered test of the deficit corollary: *if decoded symbols act as additional pilots, then starving the physical pilots should enlarge the value of a wider coding horizon*. It is also the experiment that separates the two meanings of “joint” by construction: the pilot-horizon factorial tests the code-horizon mechanism; the feedback and selection arms test continuous refinement.

##### A. Frozen design

Every arm uses one frozen two-block waveform: QPSK at a complex modem rate of 4.8828125 kHz,  $N = 2048$ ,  $N_{CP} = 64$ , rate-1/4 LDPC with check degree eight, one external frame CRC-16, two native LFM synchronization waveforms; two data blocks plus synchronization occupy 8320 samples (1.703936 s) at 20 dB added-noise SNR. The paths —

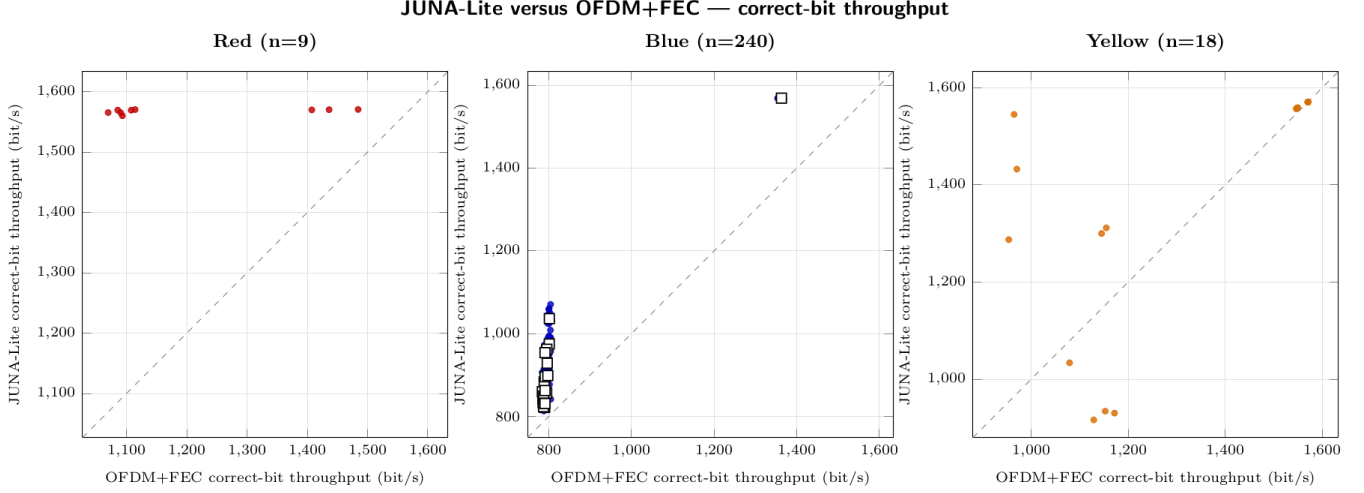


Fig. 4. Experiment A correct-bit throughput for JUNA-Lite versus OFDM + FEC. Translucent circles are all paths; outlined squares in the Blue panel are the twenty capture means. Separate family panels prevent the 240 Blue paths from hiding the nine Red and eighteen Yellow paths.

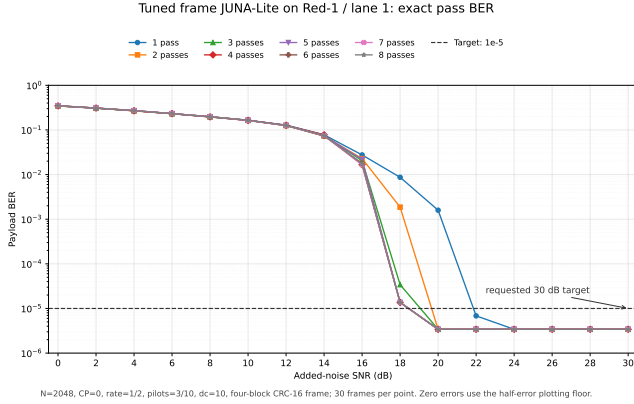


Fig. 5. Exact JUNA-Lite refinement-pass BER on the tuned Red-1 lane-1 waveform. Curves share payloads, replay snapshots, and noise within each SNR. Zero-error points are displayed at the half-error plotting floor ( $3.4153 \times 10^{-6}$ ); the archived CSV retains exact zero.

fixed before any outcome was observed — are Red-1, Blue-1, and Yellow-1, lane 1. Yellow-1 subsequently proved floor-limited and serves as a de facto control. For every family, pilot arm, and frame, the  $H_c = 1$  and  $H_c = 2$  arms share the scored payload, pilot geometry, replay interval, raw noise draw, airtime, LDPC rate and degree, and total CRC overhead. Only the code graph changes as the independent variable; encoded and received samples consequently differ, isolating graph horizon under a matched workload rather than identical samples. Four symmetric pilot spacings (3/3, 8/8, 16/16, 24/24), two matched-overhead exchange arms (5/23, 23/5; both score 1545 payload bits and expose 481 versus 480 pilot-opportunity bits), two horizons, three paths, eight receiver rows, and 180 paired frames per cell give 51,840 receiver-frame outcomes. Deterministic seeds are domain-separated by pilot arm, so outcomes are paired within an arm but *not* across arms.

The estimands, with their sign conventions collected once:

the paired horizon effect  $\Delta_H(f, p, r)$  (mean  $H_c=2$  minus  $H_c=1$  BER; negative favours the spanning graph); the pre-registered endpoint interaction  $\Gamma(f, r) = \Delta_H(f, 24/24, r) - \Delta_H(f, 3/3, r)$  (negative is the compensation direction); the allocation contrast  $A$  (negative favours outer physical pilots); and the feedback capture fraction  $\eta$  (zero reproduces frozen anchors, one reaches the genie, negative moves opposite the genie).

#### B. Primary endpoint: the interaction is not established

All three diagnostic 95% intervals for  $\Gamma$  include zero:  $[-0.005083, 0.007354]$  on Red,  $[-0.004547, 0.001988]$  on Blue, and  $[-6.771 \times 10^{-5}, 1.401 \times 10^{-5}]$  on Yellow. The point estimates disagree in sign (Table V, Fig. 6): on Red the two-block graph helps at both endpoints but helps *less* under sparse pilots ( $\Gamma = +1.135 \times 10^{-3}$  — a coding-horizon gain, but evidence *against* pilot-scarcity compensation at that operating point); on Blue the gain grows in the predicted direction ( $\Gamma = -1.280 \times 10^{-3}$ ), with coherent secondary endpoints (dense PSR  $0.55 \rightarrow 0.70$ , sparse PSR  $0.6167 \rightarrow 0.70$ ; dense GMI  $0.8088 \rightarrow 0.8491$ , sparse  $0.7551 \rightarrow 0.8193$ ); Yellow is floor-limited ( $\Gamma = -2.685 \times 10^{-5}$ ). The title question of the preregistration — does joint decoding compensate for pilot scarcity? — is therefore *not answered in the affirmative by these data*: the measured support is a path- and receiver-specific coding-horizon benefit, not a universal substitution law.

#### C. Where known information helps: the pilot exchange

The matched-overhead exchange arms hold scored payload and known-information opportunity approximately fixed while moving density between the outer physical comb and inner known message positions. The JUNA-Lite allocation contrasts are  $-0.03655$  (Red),  $-0.002287$  (Blue), and  $-9.709 \times 10^{-5}$  (Yellow) at  $H_c=1$ , with the same directions at  $H_c=2$  ( $-0.03565$ ,  $-0.001604$ ,  $-8.270 \times 10^{-5}$ ): known symbols

TABLE V

JUNA-LITE PILOT-HORIZON ENDPOINT INTERACTION.  $\Delta_{3/3}$  AND  $\Delta_{24/24}$  ARE PAIRED TWO-BLOCK HORIZON EFFECTS (NEGATIVE FAVOURS  $H_c=2$ );  $\Gamma$  NEGATIVE IS THE COMPENSATION DIRECTION. ALL THREE 95% INTERVALS FOR  $\Gamma$  INCLUDE ZERO.

| Family | $\Delta_{3/3}$ | $\Delta_{24/24}$ | $\Gamma$    |
|--------|----------------|------------------|-------------|
| Red    | -0.003283      | -0.002148        | +0.001135   |
| Blue   | -0.002241      | -0.003521        | -0.001280   |
| Yellow | 0              | -0.00002685      | -0.00002685 |

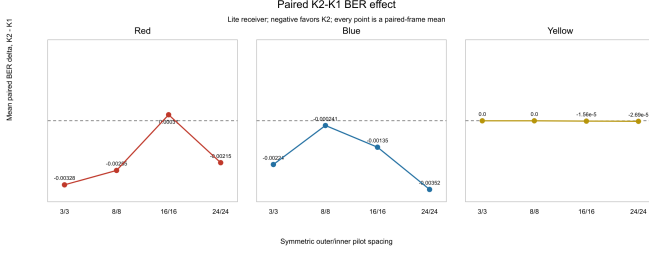


Fig. 6. Paired JUNA-Lite code-horizon effect by pilot arm (raw-data labels K1/K2 correspond to  $H_c=1/H_c=2$ ). The ordinate is  $H_c=2$  minus  $H_c=1$  BER, so lower favours the spanning graph; the registered endpoint is the sparse-minus-dense difference, not either point alone.

placed directly in the physical outer comb are much more valuable than equal known-bit opportunity inside the message on Red, and only modestly more valuable on Blue. The arms are overhead-matched but not frame-paired, so this is a descriptive allocation contrast, not a paired causal estimate. Mechanistically, an outer pilot constrains the physical observation model directly; an inner known bit acts only after coding and soft information connect it to the continuous state.

#### D. The measured bottleneck: belief-to-state conversion

The feedback ablation holds the  $(C, z)$  trajectory specification fixed and changes only the *values* supplied at fixed anchor locations: initialization (frozen), current BP posteriors (real), or transmitted symbols (genie). Relative to frozen anchors, the genie lowers the mean error count by 68.97, 14.78, and 0.02778 errors per frame on Red, Blue, and Yellow: substantial and moderate information headroom on the first two, and too little absolute headroom on Yellow for a stable normalized ratio. Real BP feedback captures 0.124% of the genie reduction on Red and 0.179% on Blue (Fig. 7); its Yellow fraction is -1.67% on a denominator of 0.02778 errors per frame and is not practically stable. CRC-gated and conditioned selection lie below the frozen reference on the normalized Red and Blue ladders because the safe gate does not preserve the small ungated mean-error reduction. The proper conclusion is not that code information is absent — the genie proves it exists — but that current BP posteriors do not convert it into sufficiently reliable pseudo-pilots. The genie is an optimistic mechanism bound, not a deployable receiver.

Consistent with this, the conditioned joint  $(C, W, z)$  solver is computationally active but endpoint-inert: across 2160 frames per family it produces a different candidate-trajectory digest from CRC-gated  $(C, z)$  on 608, 158, and 3 frames (Red, Blue, Yellow) and internally selects a conditioned update

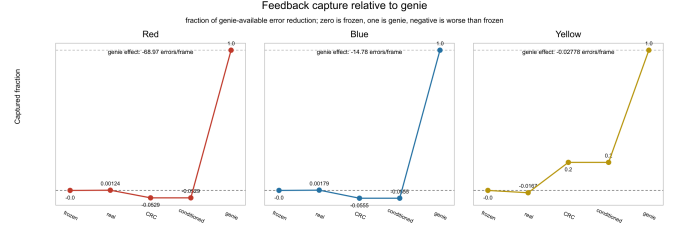


Fig. 7. Fraction of genie-available error reduction captured by feedback stages, aggregated within each family. Frozen and genie define zero and one; negative values move opposite the genie improvement.

on 12, 11, and 1 frames — yet the comparison records zero endpoint error-count changes, zero exact rescues, and zero exact breaks in all three families. Because the frame log did not persist the final decoded-payload digest for this contrast, “equal error count and equal exact-success status” is the supported statement, which is weaker than byte-identical payloads.

## XI. THE BOUNDARY FAMILY: YELLOW

### A. Matched $P = 1$ paper-geometry control

To isolate downstream receiver behavior from partial-FFT view diversity, this control forces  $P = 1$  for all five receivers and applies each channel’s own paper geometry separately to Blue-1 and Yellow-1. At  $P = 1$ , Standard OFDM and Partial-FFT have numerically equivalent front-end observations, while the five receivers remain algorithmically distinct downstream. The panels are separate replay campaigns, not a paired cross-channel comparison.

### B. Native two-LFM family stress

Yellow is a declared third-family boundary case for the deficit account under a native two-LFM protocol: six captures over lanes 1–3 at added-noise SNRs 0 : 2 : 30 dB, every one-second transmission carrying two bracketing LFM synchronization waveforms and the maximum complete OFDM blocks between them, with all nine tested receivers sharing the same payload, replay interval, noise, samples, one global frame-wide LDPC code, and CRC-16 overhead. The waveform is a nominal 6.25 kHz complex baseband at the declared 13 kHz carrier — conservative relative to the 12.5 kHz channel delay grid — so the results establish matched receiver rankings, not maximum-bandwidth Yellow throughput.

The within-family result is heterogeneous (Table VI): Yellow-1 is a ceiling case (both receivers exact on all nine frames); JUNA-Lite lowers BER on Yellow-2 without completing a frame and gives its clearest gain on Yellow-4 (0.0345  $\rightarrow$  0.0024, six exact frames rather than five); it increases BER on Yellow-3, Yellow-5, and Yellow-6. Across the complete SNR grid, OFDM + FEC has the lower equal-path BER at every tested SNR; at 30 dB JUNA-Lite completes 15/54 frames against OFDM’s 14/54 while raising equal-path BER from 0.1802 to 0.1974 — a one-frame goodput difference that is not statistically settled.

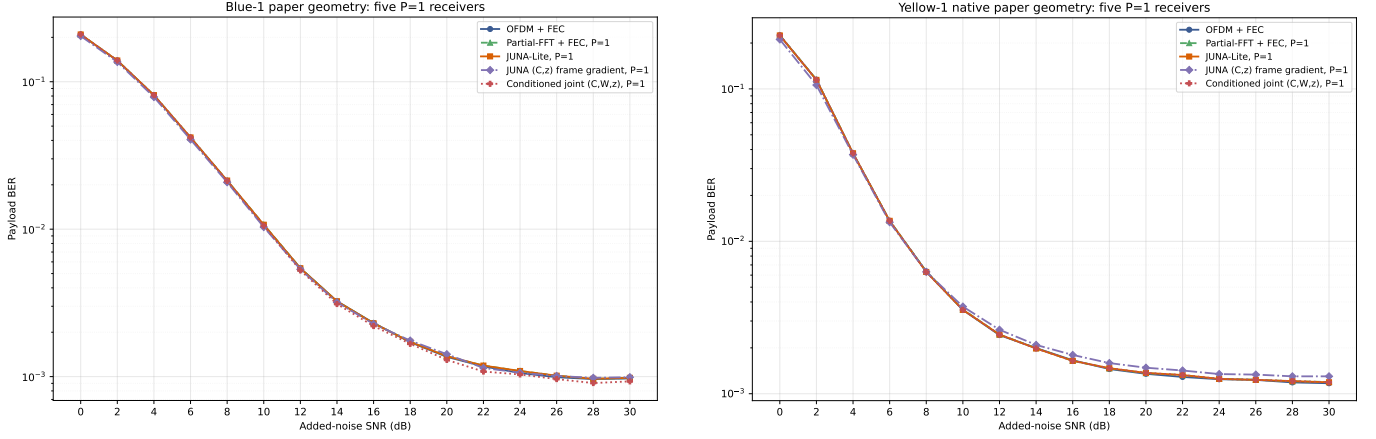


Fig. 8. Five-receiver  $P = 1$  paper-geometry controls on Blue-1 (left) and Yellow-1 (right). Blue-1 uses QPSK,  $N = 1024$ ,  $N_{CP} = 256$ , rate-1/4 LDPC, outer/inner pilot ratios 1/3 and 1/2, check degree 4, and one OFDM symbol per codeword under the Rpchanner protocol. Yellow-1 uses its native two-LFM protocol with QPSK,  $N = 2048$ ,  $N_{CP} = 64$ , rate-1/4 LDPC, outer/inner pilot ratios 1/4 and 1/4, check degree 8, and one duration-packed OFDM block. Both carry CRC-16. Each plotted point pools 1,350 receiver-matched frames within its channel: lanes 1–3, seeds 9–11, and 150 frames per lane/seed at added-noise SNR 0:2:30 dB. Payload, codeword, transmitted samples, replay interval, noise, synchronization, duration, and CRC overhead are matched across receivers within each panel. Zero-BER points are displayed at the half-error plotting floor while the CSV retains zero. These are separate replay campaigns, not a paired cross-channel comparison.

TABLE VI

YELLOW NATIVE-LFM OFDM-VERSUS-JUNA-LITE EVIDENCE AT 30 DB.  $\Delta$  IS LITE MINUS OFDM PAYLOAD BER; POSITIVE MEANS JUNA-LITE REGRESSES. EXACT DENOTES PAYLOAD-EXACT ONE-SECOND FRAMES.

| Capture         | Payload BER |        |          | Exact frames |       |
|-----------------|-------------|--------|----------|--------------|-------|
|                 | OFDM        | Lite   | $\Delta$ | OFDM         | Lite  |
| Yellow-1        | 0.0000      | 0.0000 | +0.0000  | 9/9          | 9/9   |
| Yellow-2        | 0.1764      | 0.1663 | -0.0100  | 0/9          | 0/9   |
| Yellow-3        | 0.2897      | 0.3583 | +0.0686  | 0/9          | 0/9   |
| Yellow-4        | 0.0345      | 0.0024 | -0.0321  | 5/9          | 6/9   |
| Yellow-5        | 0.3357      | 0.3632 | +0.0274  | 0/9          | 0/9   |
| Yellow-6        | 0.2447      | 0.2944 | +0.0497  | 0/9          | 0/9   |
| Equal-path mean | 0.1802      | 0.1974 | +0.0173  | 14/54        | 15/54 |

TABLE VII

YELLOW NINE-RECEIVER LADDER AT 30 DB (EQUAL-PATH MEAN OVER ALL EIGHTEEN PATHS). WHERE INVOKED, CRC-GATED AND CONDITIONED REFINEMENTS PRODUCE NO SELECTED CHANGE: THEIR OUTPUT EQUALS JUNA-LITE'S THROUGHOUT THIS CAMPAIGN.

| Receiver                        | BER      | Exact PSR | Eff. bit/s | ms/frame |
|---------------------------------|----------|-----------|------------|----------|
| OFDM + FEC                      | 0.180171 | 0.259259  | 145.4      | 23.1     |
| JUNA-Lite                       | 0.197431 | 0.277778  | 155.8      | 30.8     |
| CRC-gated ( $C, z$ )            | 0.197431 | 0.277778  | 155.8      | 106.9    |
| Conditioned joint ( $C, W, z$ ) | 0.197431 | 0.277778  | 155.8      | 89.0     |
| Frame-wide LDPC / band-RLS      | 0.207082 | 0.259259  | 145.4      | 34.5     |
| Partial-FFT + FEC               | 0.207764 | 0.277778  | 155.8      | 24.6     |
| ( $C, z$ ) frame gradient       | 0.213873 | 0.277778  | 155.8      | 140.9    |
| Profiled-Gradient               | 0.219249 | 0.259259  | 145.4      | 127.6    |
| Turbo-MAP                       | 0.226004 | 0.259259  | 145.4      | 96.1     |

The nine-receiver ladder (Table VII) sharpens the boundary: no JUNA receiver Pareto-beats OFDM + FEC on BER, exact-frame PSR, and effective payload rate, and the two refiners reproducing JUNA-Lite exactly spend roughly 3–3.5 $\times$  its decode time. Yellow therefore adds the account's most useful boundary: deployment label and raw channel difficulty alone do not predict coupling gain; waveform adequacy, initial-decision reliability, and model alignment are unmeasured

mediators that can gate whether decoder feedback is usable at all. With one noise seed and three frames per path and SNR, this campaign is descriptive rather than a powered receiver call.

## XII. SYNTHESIS: WHAT IS AND IS NOT ESTABLISHED

The evidence supports four positive statements.

- 1) *Coupling pays where the deficit is large.* Red: median +15.9% at matched waveforms (with the own-best-configuration qualifier, 1.8%), promotion on 4/9 paths at Holm-corrected  $p \leq 10^{-22}$  (the four adjusted values span  $10^{-22}$  to  $4 \times 10^{-28}$ ).
- 2) *The operating point, not the label, governs.* The same Blue family yields 0.0% median at tuned waveforms and 225/7/8 (descriptive; correlated capture-lane paths) at a deficit-inducing frozen waveform; Blue contains both a large rescue and repeatable regressions.
- 3) *The code information exists but is not being converted.* The genie removes 68.97 and 14.78 errors per frame on Red and Blue; real BP feedback captures under 0.2% of that headroom; conditioned joint selection is active but endpoint-inert.
- 4) *Iteration depth is its own lever.* 40/28/44 over 112 adjacent transitions, with regressions in nine of sixteen SNR cells — feedback needs transactional acceptance, not faith in monotonicity.

And four negative statements, which are equally part of the record.

- 1) Pilot-scarcity compensation by a wider coding horizon is *not established*: the preregistered interaction is null on all three paths.
- 2) No universal non-regression claim survives: Blue regressions and the Yellow ladder are genuine.

- 3) CRC gating and conditioned joint selection show no incremental accepted gain over JUNA-Lite anywhere in Experiment A or Yellow.
- 4) No experiment here identifies the mechanism by direct observation: channel-estimate error traces and component ablations of adaptive Lite remain future work.

The design law that survives is that coding horizon, physical pilots, feedback quality, and final acceptance are *separate levers*. The program’s present bottleneck is not the absence of code information and not optimizer effort — it is converting uncertain decoder beliefs into trustworthy continuous-state updates. Posterior calibration, reliability weighting, and feedback acceptance are therefore the implementation targets that the genie gap names precisely; adding more unconstrained continuous variables is the direction the evidence disfavours.

### XIII. LIMITATIONS

The primary Red/Blue comparisons sit at one SNR and one measurement campaign per family; search seeds vary payload and noise while replay segments are shared by design. The 240-path Blue test has nine frames per path over correlated captures and lanes and is descriptive. Experiment A uses a single common waveform and a correct-bit metric that credits erroneous bits; it diagnoses near-miss structure, not application goodput. Experiment B’s interaction intervals are diagnostic normal approximations over domain-separated arms; its exchange contrast is unpaired; its genie uses unavailable transmitted truth; and the selection-activity contrast lacks a persisted decoded-payload digest. The iteration ablation adds one path, one geometry, and one seeded schedule per SNR; its zero-error ceiling does not establish BER below  $10^{-5}$  with statistical confidence. Yellow spans sixteen SNRs but one noise seed and three frames per path and SNR at a conservative bandwidth. Family labels are strata, not samples from a population of channels. A powered, preregistered test that predicts per-path gain from an independently measured deficit remains the natural next step.

### XIV. CONCLUSION

One declared objective generates the JUNA receiver family as exact specializations, exact conditional updates, and marked approximations; one hypothesis — the estimation deficit — organizes where that family pays on measured channels; and one preregistered experiment separates the mechanisms behind it. The measured record is consistent and honest: coupling delivers systematic gains where pilots cannot resolve the channel, is neutral-to-risky where they can, and its most advertised refinements currently add computation without accepted benefit. The sharpest measured fact in the program is the gap between the genie and reality: the code knows more than the receiver uses, by a factor of five hundred. Closing that gap — calibrated posteriors, reliability-weighted anchors, transactional acceptance — is the next experiment, and the framework above is built so that when it runs, its claim will be exactly as strong as its evidence.

## APPENDIX A A RUNNABLE PATH FROM OFDM TO A CODE-AWARE RECEIVER

The main text is organized around the research claim and its measured boundaries. This appendix preserves the complementary newcomer path: begin with the public modem interface, build the partial-FFT and decoder-feedback mental model, and finish with the smallest runnable experiment. Its claims are teaching claims rather than additional measured evidence.

### A. Why Another JUNA Paper?

The journal paper answers a research question: when does a code-aware receiver improve measured underwater acoustic OFDM? A newcomer usually asks a different question: what is JUNA doing, where does the code enter, and how can I run the smallest example without learning the whole research stack first?

This folder suggests a better beginner story. It contains:

- a single runnable demo, `demo_isi_ici_awgn.jl`;
- a shared modulation interface, `Modulations.jl`;
- the implementation, `Juna.jl`;
- an LDPC wrapper, `LDPC.jl`;
- a reference write-up, `juna_doc.pdf`;
- a UnetStack/Yoda fit study in `bench/`, where Yoda (`org.arl.yoda`) is the software-defined-modem PHY that provides the usual PHYSICAL/BASEBAND layer on UnetStack/Subnero modems.

Those pieces make a natural learning ladder, and this note climbs it: start with the interface, run one channel demo, then reveal why the decoder can help the equalizer. Each section below is one rung.

### B. The One-Sentence Model

JUNA is coded OFDM with a partial-FFT equalizer and an LDPC decoder that feeds soft information back to the equalizer when Doppler smears each tone’s energy into its neighbours.

The main terms are:

| Term              | In one line                                                                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OFDM              | Orthogonal frequency-division multiplexing: send bits on many parallel tones.                                                                                                                                                                                   |
| CP                | Cyclic prefix: a guard copy prepended to each OFDM symbol to absorb echoes.                                                                                                                                                                                     |
| pilot             | A known symbol used to measure the channel; JUNA uses comb-type pilots across tones.                                                                                                                                                                            |
| partial-FFT view  | One receiver-side windowed FFT observation: samples from one short-time window of an OFDM symbol are kept, the rest is zeroed, and an $N$ -point FFT is taken. Older code may index this as a branch, but it is not an antenna, propagation path, or delay tap. |
| short-time window | The contiguous interval inside one CP-removed OFDM symbol used to form one partial-FFT view.                                                                                                                                                                    |
| ISI               | Inter-symbol interference: neighbouring OFDM symbols bleeding into one another.                                                                                                                                                                                 |
| ICI               | Inter-carrier interference: tones bleeding into one another.                                                                                                                                                                                                    |
| AWGN              | Additive white Gaussian noise: an idealized flat Gaussian noise floor; real ocean ambient noise is colored and non-Gaussian.                                                                                                                                    |
| LDPC              | Low-density parity-check: the error-correcting code JUNA decodes.                                                                                                                                                                                               |
| BP                | Belief propagation; JUNA uses a normalized min-sum approximation that also yields per-bit confidence.                                                                                                                                                           |
| LLR/logit         | Signed soft bit reliability. In robust mode, $z$ is an LLR-like bit coordinate and $\tanh(z/2)$ is the corresponding soft bit value.                                                                                                                            |
| soft              | Confidence-carrying, not just hard 0/1; unsure values shrink toward zero.                                                                                                                                                                                       |
| posterior         | Post-decoding belief about a bit or symbol after LDPC uses the parity checks.                                                                                                                                                                                   |
| RLS/LS fit        | The ridge-regularized least-squares fit used by the partial-FFT combiner; the code labels this path RLS, but it is not the recursive-LS algorithm.                                                                                                              |
| CFO               | Carrier-frequency offset. Here Doppler is a wideband time-scale $a$ ; JUNA reports it in the CFO slot as $cfo \approx (a - 1)f_c$ , the offset at the carrier.                                                                                                  |

That sentence has four parts:

- 1) OFDM carries bits on many frequency tones. Here, an OFDM symbol means one CP-protected transform block.
- 2) Multipath and Doppler stress the frame in different ways, and it helps to keep them apart. Multipath delay spread that fits inside the cyclic prefix leaves the tones separable; each tone mainly receives a complex gain. Echoes that overrun the CP spill across OFDM-symbol boundaries as inter-symbol interference. Doppler time-scales the whole waveform, which breaks subcarrier orthogonality and leaks tone energy between neighbours. The chirp sync removes the bulk time-scale; JUNA

targets the residual ICI left within each OFDM symbol.

- 3) Partial FFTs give several time-windowed views of each tone.
- 4) The LDPC decoder says which tentative symbols are trustworthy enough to reuse as soft training anchors.

### C. The Small Interface That Makes It Learnable

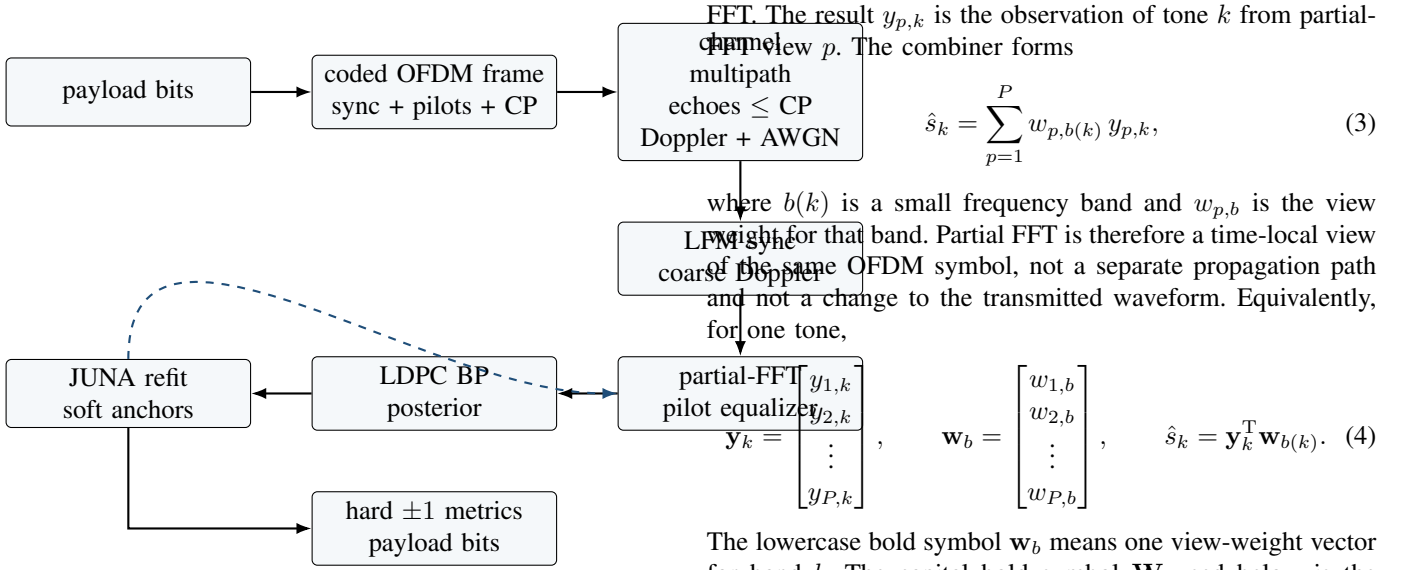
The most beginner-friendly part of the package is not a specific equation. It is the interface in `Modulations.jl`. A modem object is expected to implement the same small contract:

| Function                                     | Input idea                               | Output idea                 |
|----------------------------------------------|------------------------------------------|-----------------------------|
| <code>init(m, fc, fs)</code>                 | modem plus carrier/sample rates          | reset defaults, prep state  |
| <code>modulate(m, bits, fc, fs)</code>       | payload bits                             | complex baseband samples    |
| <code>demodulate(m, nbits, x, fc, fs)</code> | received baseband and expected bit count | bit metrics and CFO         |
| <code>bitspersymbol(m)</code>                | modem                                    | useful bits per OFDM symbol |
| <code>signallength(m, nbits, fc, fs)</code>  | desired payload size                     | sample count                |
| <code>payload_rate(m, nbits, fc, fs)</code>  | desired payload size                     | useful bits/s               |

This contract is the teaching hook. A beginner does not need to start with the full receiver. They can first understand that `modulate` turns bits into a waveform and `demodulate` turns a noisy waveform back into bit decisions. For JUNA, the returned metrics are hard-thresholded  $\pm 1$  values; the soft evidence is used inside the receiver loop, between the decoder and equalizer. One caveat is worth stating even at this level: `init` resets numeric knobs to the package's reference `red_1` modem defaults but preserves `mode` and `sync`. That is why the demo can choose a receiver mode and still call `init` safely.

### D. From OFDM to JUNA

The package can be explained as a sequence of increasingly helpful receivers.



Plain OFDM asks each carrier to stand mostly alone. Partial-FFT reception is stronger: it windows the symbol in time and combines several views of each carrier. A single full-length FFT assumes the channel holds still across the whole symbol. Under Doppler it does not, so that one coefficient is a blurred average; several FFTs, each over a shorter time window of the symbol, see less motion and give the combiner (the weighted sum of these partial-FFT views) something useful to fit.

JUNA adds one more loop. After LDPC belief propagation, each data symbol has a soft post-decoding estimate; high-confidence estimates behave most like pilots. They are not new known tones transmitted over the channel; they are decoder-supported guesses that help the receiver refit its combiner. The link to pilots is the soft-pilot trick. A pilot works because the receiver already knows the transmitted symbol, so the distortion on that tone is a direct channel measurement. A data symbol that the LDPC decoder returns with high confidence is almost as good as known, so JUNA treats it as a soft pilot and refits the equalizer against this enlarged set of references. In the lite path, uncertainty is handled by filtering the anchor set: low-confidence data tones are not reused as training anchors. In the robust path, the update weights training by decoder confidence, so unsure bits contribute less.

In communications terms this is an iterative decode-and-re-estimate loop, closer to soft decision-directed channel estimation than to strict turbo equalization. Rather than exchanging extrinsic LLRs with a soft-input soft-output equalizer, JUNA reuses posterior symbol means, dominated by the confident tones, as soft pilots to re-fit its least-squares combiner.

#### E. The Cost Behind Lite and Robust

The code and older notes sometimes call the partial-FFT index a *branch*, but that word is overloaded in communications: it can mean an antenna, a diversity path, or a channel tap. Here we use *partial-FFT view* for each receiver-side windowed FFT observation. After the cyclic prefix is removed, the receiver cuts one OFDM symbol into  $P$  nearly equal short-time windows. For each window it keeps that rectangular time interval, zeros the rest of the symbol, and takes an  $N$ -point

The lowercase bold symbol  $\mathbf{w}_b$  means one view-weight vector for band  $b$ . The capital bold symbol  $\mathbf{W}$  used below is the whole collection of those vectors, one per band:  $\mathbf{W} = \{\mathbf{w}_1, \dots, \mathbf{w}_B\}$ . They are the same kind of combiner weight, but at different scopes.

Both `:lite` and `:robust` start from the same simple combiner-fit problem. Suppose a set of tones  $T_b$  in band  $b$  has target symbols  $d_k$ . For pilots,  $d_k$  is known. After LDPC decoding,  $d_k$  can also be a posterior-mean data symbol. In the lite update, low-confidence data tones are left out of the anchor set; only tones above the confidence threshold, up to the anchor cap, are reused as soft training targets. The combiner weights for that band are fit by

$$\min_{\mathbf{w}_b} \sum_{k \in T_b} \left| \sum_{p=1}^P w_{p,b} y_{p,k} - d_k \right|^2 + \lambda_{\text{ridge}} \|\mathbf{w}_b\|_2^2. \quad (5)$$

If  $T_b = \{k_1, \dots, k_m\}$ , the matrix form stacks the selected tones as

$$Y_b = \begin{bmatrix} \mathbf{y}_{k_1}^T \\ \mathbf{y}_{k_2}^T \\ \vdots \\ \mathbf{y}_{k_m}^T \end{bmatrix} = \begin{bmatrix} y_{1,k_1} & y_{2,k_1} & \cdots & y_{P,k_1} \\ y_{1,k_2} & y_{2,k_2} & \cdots & y_{P,k_2} \\ \vdots & \vdots & \ddots & \vdots \\ y_{1,k_m} & y_{2,k_m} & \cdots & y_{P,k_m} \end{bmatrix}, \quad \mathbf{d}_b = \begin{bmatrix} d_{k_1} \\ d_{k_2} \\ \vdots \\ d_{k_m} \end{bmatrix}. \quad (6)$$

Thus the fit is  $\|Y_b \mathbf{w}_b - \mathbf{d}_b\|_2^2 + \lambda_{\text{ridge}} \|\mathbf{w}_b\|_2^2$ . Differentiating with respect to  $\mathbf{w}_b^*$  and setting the result to zero gives

$$(Y_b^H Y_b + \lambda_{\text{ridge}} I) \mathbf{w}_b = Y_b^H \mathbf{d}_b, \quad (7)$$

which is the small normal equation solved in the code. That is the derivation of the lite update. The receiver first seeds a partial-FFT plus FEC candidate; if that candidate is already valid, it keeps it and skips the refit. Otherwise it freezes the decoder posterior means as soft targets, chooses confident data tones as additional anchors, refits the least-squares combiner, decodes again, and keeps the better candidate. Lite can therefore be read as one decision-directed block update of the combiner with the symbol beliefs held fixed. It is not a gradient descent on every receiver variable.

The robust mode is the gradient version of a richer, but still reduced, cost. It keeps the full combiner collection  $\mathbf{W}$  as a

**Algorithm 1** JUNA lite posterior-anchor refit

**Require:** Partial-FFT observations  $y_{p,k}$ , pilots, LDPC decoder, anchor cap  $A_{\max}$

- 1: Fit pilot-only combiners  $\mathbf{W}_0$  by ridge/RLS least squares
- 2: Decode the partial-FFT+FEC seed and record its validity, syndrome, score, and posterior confidence
- 3: **if** the seed is already acceptable **then**
- 4:     **return** the seed candidate
- 5: **end if**
- 6: Freeze the decoder posterior means  $\tilde{d}_k$  and confidences  $c_k$
- 7: **for** each frequency band  $b$  **do**
- 8:     Select at most  $A_{\max}$  confident data tones in band  $b$
- 9:     Stack pilots and selected data anchors into  $Y_b$  and  $\mathbf{d}_b$
- 10:     Solve  $(Y_b^H Y_b + \lambda_{\text{ridge}} I) \mathbf{w}_b = Y_b^H \mathbf{d}_b$
- 11: **end for**
- 12: Equalize with the refit combiner collection  $\mathbf{W}$ , then LDPC-decode again
- 13: **return** the better candidate under validity, syndrome, score, and posterior-confidence tests

variable and also lets  $z$ , a vector of LLR-like bit coordinates, move. These coordinates define soft QPSK symbols through

$$S_t(z) = \frac{1}{\sqrt{2}} \left[ \tanh\left(\frac{z_{2t-1}}{2}\right) + j \tanh\left(\frac{z_{2t}}{2}\right) \right], \quad (8)$$

with the second term omitted at the end if the codeword length is odd. In the data-tie term,  $k_t$  is the tone that carries data symbol  $t$ . The channel response, Doppler time-scale, and timing are not variables in this objective; they have already been folded into the partial-FFT observations  $y_{p,k}$ , the synchronization step, and the pilot fit  $\mathbf{W}_0$ . Thus  $J$  optimizes only the engineered receiver variables  $\mathbf{W}$  and  $z$ , not a joint physical-channel likelihood. The LDPC code constraints are fixed; they enter as a parity penalty on  $z$ , and they influence  $\mathbf{W}$  only through the data-tie term that couples  $\hat{s}_{k_t}(\mathbf{W})$  to  $S_t(z)$ . For readability, the coefficients below absorb the implementation's per-count normalizations over pilots, data tones, checks, and variables:

$$J(\mathbf{W}, z) = \alpha \sum_{\text{pilots}} |\hat{s}_k(\mathbf{W}) - p_k|^2 + \beta \sum_{\text{data } t} c_t |\hat{s}_{k_t}(\mathbf{W}) - S_t(z)|^2 + \lambda \sum_{\text{checks}} \left( 1 - \prod_{i \in \text{check}} \tanh(z_i/2) \right)^2 + \eta \|\mathbf{W} - \mathbf{W}_0\|^2 + \mu \|z - z_0\|^2 + \gamma \|z\|^2, \quad (9)$$

The first term keeps known pilots matched. The second ties equalized data tones to the current soft symbols, weighted by decoder confidence  $c_t$ , computed once from the BP-seed posterior  $z_0$  and then held fixed during the Adam updates. The third is a smooth LDPC parity-check penalty:  $\tanh(z_i/2)$  is the soft  $\pm 1$  value of bit  $i$ , and a satisfied parity check has product  $+1$ , so this term penalizes a soft check product that drifts away from satisfaction. The  $\eta$  and  $\mu$  terms keep the combiner near the pilot fit  $\mathbf{W}_0$  and the LLR coordinates near the BP seed  $z_0$ . The final  $\gamma \|z\|^2$  term is a plain ridge that shrinks undecided LLR coordinates toward zero. Together these regularizers stop the optimizer from drifting far from the pilot-fit and BP-seed starting point. Unlike lite, robust

**Algorithm 2** JUNA robust reduced gradient refinement

**Require:** Seed  $(\mathbf{W}_0, z_0)$ , fixed confidences  $c_t$ , fixed LDPC checks, step budget  $G$

- 1: Initialize  $\mathbf{W} \leftarrow \mathbf{W}_0$ ,  $z \leftarrow z_0$ , and best candidate from the seed decode
- 2: **for**  $g = 1, \dots, G$  **do**
- 3:     Form soft symbols  $S_t(z)$
- 4:     Evaluate  $J(\mathbf{W}, z)$  from Eq. (7)
- 5:     Compute gradients  $\nabla_{\mathbf{W}} J$  and  $\nabla_z J$
- 6:     Take one Adam step on  $\mathbf{W}$  and  $z$
- 7:     Equalize with the updated  $\mathbf{W}$ , map  $z$  to LLRs, and LDPC-decode a checkpoint
- 8:     Keep the best checkpoint under validity, syndrome, score, and posterior-confidence tests
- 9: **end for**
- 10: **return** the best decoded candidate

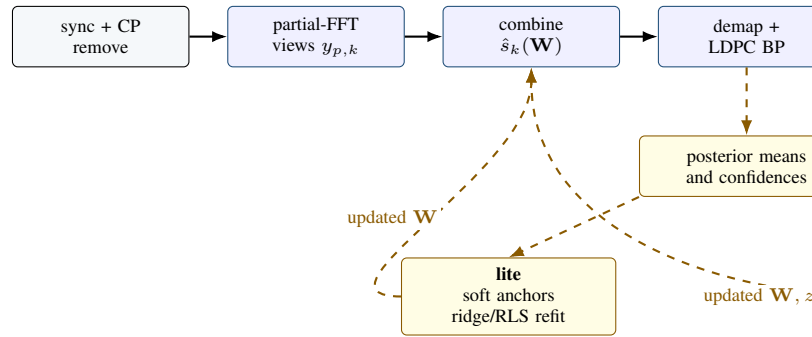


Fig. 9. Pictorial view of the JUNA feedback loop. The top row is the seed receiver path. The dashed loop is the decoder-to-equalizer feedback inspired by the slide deck: LDPC posteriors either become soft anchors for a lite least-squares refit or drive a robust gradient step, after which the receiver equalizes and decodes another candidate.

has no closed-form update;  $J(\mathbf{W}, z)$  is a hand-built reduced objective, and the code descends its gradients with Adam. The receiver decodes a candidate at each step and keeps the best decoded candidate under validity, syndrome, score, and mean-absolute-posterior tests.

So the hierarchy is:

- `:lite`: closed-form least-squares refit with posterior symbols held fixed as soft targets;
- `:robust`: gradient descent on the reduced  $\mathbf{W}, z$  objective, where combiner weights and soft symbol LLR coordinates move together.

Calling `:robust` “complete” is only safe in this limited sense: it is the complete implemented cost for the reduced  $\mathbf{W}, z$  receiver path, meaning the variables it actually moves are the combiner weights and the soft code-symbol LLR coordinates. It is not a full physical-channel likelihood, not a timing/acquisition optimizer, and not a joint estimator over passband propagation, Doppler trajectory, carrier phase, sample timing, synchronization, and codeword.

### F. The Local Demo as a First Experiment

The first step is a one-time environment setup:

```
julia --project=. -e 'using Pkg; Pkg.instantiate()'
```

Run these commands from inside the `Juna/` folder. That environment step is needed once on a fresh checkout. The LDPC helper tools `make-ldpc`, `make-gen`, and `print-pchk` must also be available on `PATH` or beside `LDPC.jl`; they are platform-specific binaries and are not committed in this folder. Without them, code construction fails with a `could not spawn make-ldpc error`. With the environment and LDPC tools in place, the first actual experiment is:

```
julia --project=. demo_isi_ici_awgn.jl
```

In the current folder, one seeded run prints:

```
channel: 4-tap multipath (max delay 142 <= CP, 256)
        + Doppler a=1.0015 + AWGN 20 dB
lite    : a_est=1.00145 (cfo=+34.88 Hz) BER = 0/1024 = 0.000e+00
robust  : a_est=1.00145 (cfo=+34.88 Hz) BER = 0/1024 = 0.000e+00
```

*Transcript note: the program prints the channel description on one line; it is wrapped here to fit the page. The program also prints a Unicode less-than-or-equal sign where this transcript uses <=.*

This small printout teaches several ideas at once:

- **The echoes fit inside the CP.** The 142-sample maximum excess delay is shorter than the 256-sample cyclic prefix, so the synthetic echo tail does not spill into the next block; each tone is mainly left with a frequency-selective complex gain. Because the sync removes the bulk time-scale, the residual Doppler-induced ICI in this run is small and the pilot equalizer, combiner, and LDPC clean it up comfortably at 20 dB.
- **Doppler is present.** The waveform is time-scaled by  $a=1.0015$ ; the sync estimates it as about 1.00145. The receiver reports this through the `cfo` field as  $(a_{\text{est}} - 1)f_c$ , so the rounded value  $(1.00145 - 1) \times 24 \text{ kHz}$  gives about +34.8 Hz; the transcript's +34.88 Hz uses the unrounded estimate. Here the positive sign follows the code's time-scale convention, positive for  $a > 1$ , not a signed physical carrier shift. The underlying time-scale shifts each tone by an amount proportional to its frequency, so a single carrier offset cannot fully describe it. At a 1500 m/s sound speed, that scale is roughly 2.2 m/s of relative motion.
- **The channel is at a favorable noise level.** AWGN is added at a 20 dB signal-to-noise ratio; higher dB means less noise, and this favorable level is part of why both modes decode this frame cleanly.
- **Both receiver modes succeed on this realization.** The demo is not a proof of universal performance; it is a compact sanity check.

### G. Lite and Robust Modes

The same modulation object has two receiver modes:

| Mode                 | Mechanism                                           | Teaching interpretation                |
|----------------------|-----------------------------------------------------|----------------------------------------|
| <code>:lite</code>   | post-decoding anchor least-squares refit            | fast decoder feedback via soft anchors |
| <code>:robust</code> | reduced gradient over combiner and symbol estimates | higher-cost path for harder ICI        |

For a newcomer, the key idea is not the specific solver; it is that both modes use decoder information before giving up on the equalizer. `:lite` is the first one to teach because it has the cleanest mental model: trusted decoded symbols become soft anchors.

*H. What the UnetStack Bench Adds*  
The `bench/` folder contributes a second beginner paper angle: JUNA is not just an isolated simulation. It can be explained as a modem that fits a larger acoustic networking stack.

Here “Yoda-style OFDM” concretely means `bench/ofdm_musique.jl`, a Julia clean-room reference for the Yoda ofdm waveform. Deployed Yoda itself is a closed Java/native stack that this folder does not run against.

The file `bench/explain_juna_yoda_sim.jl` teaches three boundaries:

- 1) the `musique` OFDM reference and JUNA can share the same Julia modulation API;
- 2) JUNA can ride a Yoda-shaped baseband path of interleaved complex samples;
- 3) sharing an interface does not mean the Yoda-style OFDM receiver in the `musique` reference can decode JUNA waveforms, or vice versa.

That is a useful newbie distinction. “Compatible with the stack” means the objects have the same service shape. It does not mean the two waveforms are bit-compatible.

### I. Honest Limits

Six facts this demo does not establish:

- The demo does not prove measured-channel performance.
- It does not show that JUNA is the same waveform as Yoda OFDM.
- It does not show that the robust mode always beats the lite mode.
- It does not net out pilot, FEC, CP, or sync overhead.
- Its 20 dB number is a per-sample complex-baseband SNR, not an  $E_b/N_0$  point for a rate-matched BER curve.
- It runs at complex baseband and starts from an isolated captured frame; it skips passband up/down-conversion, AGC, fine timing, and detecting the frame inside a continuous noisy stream. The LFM sync provides coarse frame alignment and Doppler, not a full deployed receiver front end.

The right claim is more modest and more useful: this folder gives a runnable, minimal ladder for understanding JUNA from the interface upward.

### *J. Takeaway*

The strongest beginner explanation is not “JUNA is a better OFDM.” It is:

JUNA is what happens when an OFDM receiver stops treating the decoder as the last step and starts treating it as a source of soft evidence for the equalizer.

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